ZOO603-T

VU Expert B.S Zoologists
(All Semesters)

FINALTERM SUBJECTIVE

SPRING 2024

By

★ AlifExpertZoologist ★

VU EXPERT ZOOLOGISTS

IMPORTANT QUESTIONS

1.What is Common Carp? Write its characteristics.

Common carp ***Cyprinus carpio*** was introduced from Thailand to Pakistan in 1964 for the purpose of aquaculture. Due to its high tolerance to temperature and turbidity, and prolific pond breeding habit, it was established promptly in most of natural inland waters, including rivers, lakes, streams, canals, wetlands and even village ponds of the country. Although common carp became one of the most abundant cyprinid species in inland waters and important food fish in Pakistan, its impact is not well documented.

Fish farming of common carp has been carried out in Pakistan since 1970; initially it grew slowly but now it is playing an important role in the economy of the country by employing more than 400,000 people. Nowadays, farming of freshwater carps is present throughout Pakistan, especially in the provinces of Punjab and Sindh.

2. What is the Law of Tolerance?

"The presence and success of an organism depend upon the completeness of a complex of conditions. Absence or failure of an organism can be controlled by the qualitative or quantitative. Deficiency or excess with respect to any one of several factors which may approach the limits of tolerance for that organism."

For example: The ability to exist and thrive may depend on such physical factors as dissolved oxygen, temperature, salinity, pH, etc ."Steno" refers to species with narrow tolerances (e.g..

Stenothermal= Narrow temperature tolerance) “eury” refer to those with wide tolerance (e.g. euryhaline = wide salinity tolerance)

3. Name the causative agent of Furunculosis and Abdominal dropsy. OR What is a bacterial disease?

Bacterial Diseases:

A **bacterial disease** in fish is an illness caused by the presence of harmful bacteria that infect the fish, leading to various symptoms such as lesions, swelling, or organ failure. These diseases can be highly contagious and affect the health of fish populations.

Causative Agents of Specific Bacterial Diseases:

1. **Abdominal dropsy of Carps** is caused by **Aeromonas punctuata**.
2. **Furunculosis of Salmon and Trout** is caused by **Aeromonas salmonicida**.

4. What is Biological Rhythm?

We are strongly controlled by internal (endogenous) circadian rhythms affecting temperature, physiology and behavior etc. Physiology and behavior are strongly controlled by such rhythms, with periods of a year, a lunar month, a day or a tidal cycle. The reproductive biology and feeding patterns are intimately linked to such rhythms.

Example include vertical feeding migrations from deep water towards the surface to feed on plankton, annual migrations from feeding to breeding grounds, synchrony of breeding on spring high tides and movement from the bottom to the middle of the water column on high and low tides.

5. What do you know about the caudal fins of Bonyfish? What are its types?

The Caudal fin or Tail fin is the primary fin of the fish that gives it forward propulsion and speed. For **fish identification**, the type of Caudal fin is used to distinguish one type from another.

The main types of Caudal fins found in bony fish are:

- Homocercal shape tail
- The Heterocercal tail
- Round tail
- Truncated (triangular) tail
- Lunate (moon shaped) tail
- Forked tail

1. Homocercal shape tail: It is the most common caudal fin shape, shared by most fishes. It includes square, truncate, slightly forked and deeply forked types.
2. The heterocercal tail: Possessed by only a few primitive fishes such as sturgeon, paddlefish and sharks.

3. Rounded tail: Slow swimming, accelerating, and maneuvering.
4. Truncated (triangular) tail: Turning quickly
5. Lunate (moon shaped) tail: Continuous long distance swimming

6. Forked tail: Rapid swimming, somewhat sustained with bursts of speed.

6. What is Chimaera? Write its characteristics.

Chimaera, (subclass Holocephali), any of numerous cartilaginous fishes related to sharks and rays in the class Chondrichthyes but separated from them as the subclass (or sometimes class) Holocephali. Like sharks and rays, chimaeras have cartilaginous skeletons, and the males possess external reproductive organs (claspers) derived from the pelvic fins and used to introduce sperm into the body of the female.

Unlike sharks and rays, chimaeras have a single external gill opening, covered by a flap as in the bony fishes, on each side of the body. Male chimaeras, unique among fishes, also possess a supplemental clasping organ, the tentaculum, on the forehead and in front of each pelvic fin. Chimaeras are tapered fishes with large pectoral and pelvic fins, large eyes, and two dorsal fins, the first preceded by a sharp spine. They are weak swimmers and are delicate when caught, dying quickly out of water. Their food consists of small fishes and invertebrates. Chimaeras are edible and are sold as food in some areas.

7. What is Fugu poisoning?

Pufferfish poisoning:

Tetrodotoxic poisoning occurs following the ingestion of pufferfish, boxfish or porcupine fish. This is also known as **puffer or fugu poisoning**. The poisoning is due to toxin known as tetrodotoxin which is concentrated in the internal organs such as liver, ovary and intestines. The skin also contains some glands which secrete these toxins. Usualy the body muscles are free of the toxin.

8. What is Pisciculture?

Pisciculture, also known as fish farming or culture fisheries, is the practice of breeding, rearing, and harvesting fish in controlled environments such as ponds, tanks, or small water bodies. It involves a systematic process where fish seeds (eggs or fry) are sown, nurtured, and raised under monitored conditions. Proper care is taken to maintain optimal water quality, provide suitable feed, and protect the fish from diseases or predators. Once the fish reach table size or marketable size, they are harvested for consumption or commercial purposes. This sustainable practice helps meet the growing demand for fish and reduces pressure on wild fish populations.

9. What are Lobsters?

Lobsters are one of the most valuable and highly priced crustaceans as well as an important export commodity. The massive head bears five pairs of walking legs; the abdomen is well developed, muscular and bears five pairs of swimmerets. Lobsters are solitary animals that crawl on the sea floor and hide under rocks.

Among the 12 species recorded from Indian coastal waters, only four species of Spiny Lobsters (Panulirus homarus, P. ornatus, P. polyphagus, P. versicolor) and one species of Slipper Lobster (Thenus orinetalis) are commercially important. Growth is slow, most lobsters take more than 3 years to attain maturation and they have a prolonged life-cycle.

Grow-out and fattening of lobsters is commonly done in indoor tanks and cages to fetch better price in market.

10. Write three functions of ecosystems.

Functions of Ecosystem:

The functions of the ecosystem are as follows:

- It regulates the essential ecological processes, supports life systems and renders stability.
- It is also responsible for the cycling of nutrients between biotic and abiotic components.
- It maintains a balance among the various trophic levels in the ecosystem.
- It cycles the minerals through the biosphere.
- The abiotic components help in the synthesis of organic components that involve the exchange of energy.

11. Write the purpose of aquaculture.

Purpose of Aquaculture:

- Utilization of available natural water resources
- To increase production for per capita consumption and income
- Ornamental purpose
- Sports and game purpose.
- Upliftment of socioeconomic status of people
- Create employment opportunities.
- Utilization of byproducts like fish, liver oil, fish protein concentrate, etc.

12. Write the characteristics of aquaculture.

1. **High Productivity:** In comparison to agriculture or veterinary practices, aquaculture has been found to be more productive.
2. **Integrated Farming:** Aquaculture with agriculture, horticulture or animal husbandary is found to be more profitable,
3. **Recycling:** Aquaculture gives efficient means of recycling domestic and/or agriculture wastes and thus, helps in protecting our environment.
4. **Rural Development:** Aquaculture helps to integrate rural development by generating employ-ment opportunities and would thus help to arrest the exodus of population from rural to urban areas.
5. **Intensive Fish Culture:** In aquaculture practice, owing to the fishes three-dimensional utilization of the water body can be crowded more closely (200/ m3) and grown through water recirculation system. This gives high yield of about 25 t/ha/year.
6. **Earning Foreign Exchange:** Commercially important items, such as ornamental fishes, Artemia cysts, prawns, lobsters, crabs, frog legs, etc., produced through aquaculture are highly valued and can earn good foreign exchange.
7. **Ranching:** Fish seeds and larvae of economically important fishes of capture fisheries are artificially recruited in fish hatcheries through aquaculture, called ranching or aquarange farming. It is commonly known as **“culture based capture fishery”** and

involves restocking of the wild stock for harvesting. This process has replenished the dwindling stock of rivers and seas.

8. **Replenishment of Coral Reefs:** Coral reefs that have been destroyed naturally or anthropogenically are being replenished through construction of artificial coral reef.
9. **Creating Leasure-Time Activities:** Leasure- time activities can be maintained through sport fishing and creating home and public aquaria.
10. **Mariculture and Fisheries Enhancement:** In many reef areas the use of mariculture is fast growing. It provides an alternative source of income and employment in coastal areas. Replenishment schemes of species that has been greatly diminished has been undertaken in certain areas. This has been useful for returning species such as giant clams and trochus to reefs where they have been largely or completely exterminated by overfishing.
11. **Promoting Agro-Industrial Development: Aquaculture can promote agro-industrial development through:**
    i. Processing and marketing of fishery products, feeds and equipments for aquaculture,
    ii. Seaweed culture for the production of marine celluloids, and
    iii. Pearl oyster culture, etc.

13. Write the distinguishing features of Osteichthyes.

Class Osteichthyes:

With more than 29,000 species of bony fishes, it is the largest class of vertebrates. It includes both marine and freshwater fishes, most of the commercially used fishes are included in this class. The size ranges from less than 8 mm in *Paedocypris progenetica*, which is also the smallest known vertebrate to 4 m and weigh about 1500 kg in the ocean sunfish (*Mola mola*). Their endoskeleton is made up of bones. Bony skeleton. Operculum cover gills opening. 20,000 species.

Present in Freshwater and saltwater. Pneumatic sacs functions as lungs or swim bladder. Bone in their skeleton, bony operculum covering the gills openings, lungs or swim bladder. Lateral line system. **Fins**: Dorsal, pectoral, caudal, pelvic, anal. Paired and median fins are present, which are supported by long rays of cartilage or bone. Fleshy lobed fins are present in sarcopterygians. These types of fins are supported by bones having joints. These are adapted to live at the bottom of the sea. First fossils of bony fishes are from late Silurian deposits

Example: Bony fishes

14. What is the function of lateral line system?

Lateral line system:

All of the primarily aquatic vertebrates—cyclostomes (e.g., lampreys), fish, and amphibians—have in their outer skin (epidermis) special mechanoreceptors called lateral-line organs. These organs are sensitive to minute, local water displacements, particularly those produced by other animals moving in the water.

The **lateral line system** consists of **100** or more sensory organs (neuromasts) that are typically arranged in **lines** on or just under the skin of the head and body. It is absent from all

reptiles, birds, and mammals, even those that are aquatic (such as turtles, dolphins, and whales). In the head, the lateral line canal is separated into three canals, one passes forward and above the eye, another forward and below the eye and the other downward and below the jaw. These three canals have numerous pores and together with the lateral line canal, make the lateral line system.

15. Name any three international shrimp species.

Some of the commercially important marine shrimps are Tiger Prawn (Penaeus monodon), Indian White Prawn (Penaeus indicus), Pink Shrimp (Metapenaeus dobsoni), Brown Shrimp (Metapenaeus monoceros), etc. For Aquaculture purpose, the exotic Pacific White Shrimp (Litopenaeus vannamei) has been introduced into India and is being farmed extensively on a commercial-scale in low-saline or brackishwater ponds in Gujarat, Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Odisha, West Bengal etc

16. Write the characteristics of fish.

General characteristics:

- Fish are aquatic, gill-bearing animals that lack limbs with digits.
- Most fish are cold-blooded, allowing their body temperatures to vary as ambient temperatures change.
- Fish share with vertebrate a basic body plan of bilateral symmetry.
- Their body is divisible into head, trunk and tail. Neck is absent.
- It has a spindle shaped body. It is helpful in swimming.
- Respiration is by gills. Gills are the extensions of the pharynx.
- On the head a pair of eyes is present.
- On the lateral sides of the body LATERAL LINE SENSE **organs** are present. They detect the pressure changes of water.
- The digestive system is well developed.
- 10 pairs of cranial nerves are present.
- Kidneys are mesonephric.
- Urinary bladder is absent
- The skeleton of some fishes is made by **cartilage**. They are called cartilage fishes. In some fishes the skeleton is made by bone. They are called bony fishes.
- Sexes are separate.
- Many fishes are oviparous. Some fishes are **viviparous**.

17. Feeding of pf herring like fish?

Herring-like fish, such as herring, sardines, and anchovies, are typically planktivores or filter feeders, feeding on:

- Phytoplankton (microscopic plants)
- Zooplankton (microscopic animals)
- Small crustaceans (such as krill or copepods)
- Small fish larvae

They use their small teeth and gill rakers to filter the water and trap their food. Some species may also feed on detritus or algae.

18. Describe the feeding behavior of PF-herring fish.

Filter feeding: using their gill rakers to strain the water and capture small particles
Surface feeding: feeding on the surface of the water, often in large schoolsl
Mid-water feeding: feeding in the middle of the water column, often on small fish or crustaceans.

19. What are jugular fins?

When pelvic fins are below the pectoral fins, they termed thoracic. When a thoracic pelvic fin is attached under gills, it may called **jugular**.

OR

Jugular fins refer to pelvic fins that are positioned further forward on a fish's body, located beneath or in front of the pectoral fins, often near the gills. The pelvic fins are termed "**thoracic**" when located beneath the pectoral fins. However, when these thoracic pelvic fins are attached near or under the gill region, they are specifically referred to as "**jugular fins**." This adaptation is more common in bottom-dwelling or benthic fish species. Some fish with jugular fins include species like dragonets and certain members of the cod family. This evolutionary feature is thought to help these fish in foraging or navigating their environments effectively.

20. What is the primary function of the bioelectric organ? or what is the electric organ in fish?

Bioelectric organ, system of tissues specialized for the production and use of electrical power in a living organism.

•Well developed in a wide variety of fishes, both marine and freshwater, indicating an early evolutionary development, bioelectric organs probably represent a specialization of a common bioelectrical capacity of all living cells.

• (Various other tissues and organs also possess the capacity to produce electricity—the skin of frogs and the heart, brain, and eye of higher animals including humans.)
•In more than 200 **fish** species, the bioelectric **organ** is involved in self-defense or hunting.

•The torpedo, or **electric ray**, and the **electric eel** have especially powerful electric organs, which they apparently use to immobilize or kill prey.

21. Write the economic importance of shrimps.

Economic Importance of Shrimps

Shrimps hold significant economic value, particularly in the seafood and aquaculture industries. Some of the key commercially important marine shrimp species include Tiger Prawn (***Penaeus monodon***), Indian White Prawn (***Penaeus indicus***), Pink Shrimp

(***Metapenaeus dobsoni***), and Brown Shrimp (***Metapenaeus monoceros***). These species are widely harvested and exported, contributing substantially to the economy.

Additionally, the exotic Pacific White Shrimp (***Litopenaeus vannamei***) has been introduced to India for aquaculture purposes. It is now extensively farmed on a large commercial scale in low-saline or brackish water ponds across states like Gujarat, Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Odisha, and West Bengal. This farming has greatly boosted shrimp production, supporting livelihoods and enhancing export revenue.

22. What is brackish water? Is it related to aquaculture or ecosystems?

Brackish Water Aquaculture:

Brackish water areas are rich in oxygen and plankton. Brackish water organisms such as milkfish (***Chanos chanos***), Lates calcarifer, Mugil sp. prawns, mussels, etc. can be cultured in various ways such as in tidal ponds, pens, cages or by rack, raft or rope culture (mainly in case of oysters). Mangrove crab, Scylla serrata, are also cultured in the shallow brackish water regions by constructing bamboo cages.

23. Differentiate between bays and estuaries.

Bay:

A bay is a body of water partially enclosed by land, connected to the open sea, with saline water. It typically has a wide mouth and is often circular or oval in shape, with water that is usually saline and similar to the open ocean. ***Examples*** of bays include the Bay of Bengal, San Francisco Bay, and Hudson Bay.

Estuary:

An estuary is a coastal body of water where freshwater from rivers mixes with saltwater from the ocean, creating brackish water with a mixture of fresh and saltwater. Characterized by a narrow mouth and is more saline than freshwater but less saline than seawater. ***Examples*** of estuaries include the Chesapeake Bay, San Francisco Estuary, and Amazon River Estuary.

24. What characters are included in fish morphology?

Fish morphology refers to the study of the form and structure of fish, encompassing a wide variety of species and specialized terminology for their physical traits. It plays a vital role in understanding the behavior, habitat preferences, and ecological niches of different fish species. Morphology also provides insight into a fish's evolutionary history and its ability to adapt to changing conditions.

The main characteristics of fish morphology include:

- Body shape
- Mouth type
- Mouth location
- Scales
- Gills

- Fin type
- Barbels
- Lateral line system

25. Write the adaptations of gills in fishes.

Adaptation of gills for efficient gas exchange are:

1. Thin (Flattened cell): Short diffusion distance

2. Permeable membranes: Allow gases to diffuse

3. Good vascular supply: Maintain concentration gradient

4. Large surface area: Allow more diffusion of gases.

26. What is the blue and green revolution?

"BLUE REVOLUTION" (fish production) on the same lines
A period of rapid growth in aquaculture, achieved through the development of new technologies, feeds, and management practices, aimed at increasing seafood production.

'GREEN REVOLUTION' (for producing enough food for all).

A period of significant increase in agricultural productivity, achieved through the use of high-yielding crop varieties, irrigation, fertilizers, and pesticides.

27. Differentiate between lotic and lentic freshwater ecosystems.

Types of Freshwater Ecosystem

The freshwater ecosystem is mainly divided into three types based on its region – Lotic, lentic, and wetland freshwater ecosystem.

Lotic Freshwater Ecosystem:

In simple terms, the water bodies moving in one direction is known as a lotic freshwater ecosystem. Rivers and streams are common examples of lotic ecosystems. There are many rivers and stream that flows from their origin and ultimately meets with other water channels or oceans at its mouth. Lotic freshwater travel through different locations from its source to mouth.

Lentic Freshwater Ecosystem:

An aquatic ecosystem within stagnant or still water like ponds and lakes is known as Lentic Freshwater Ecosystem. Lentic ecosystem found in various sizes ranging from a few square meters to thousands of square km. Some ponds last just for a few months as these are seasonal like sessile pools. On the other hand, lakes may exist for many years. Lentic ecosystem, i.e Ponds and lakes, support a very limited number of species.

28. Differentiate between Actinopterygii and Sarcopterygii.

Subclass Actinopterygii:

Ray-finned fish. Their fins lack muscular lobes that is why called as ray-finned fishes. They usually possess swim bladders, gas filled sacs along dorsal wall of the cavity that regulate buoyancy. Paired fins supported by dermal rays. Base of fins is not muscular. Tail fins almost equal to upper and lower lobes. Blind olfactory sacs.

Examples: Ray fined fishes

Subclass Sarcoptergyii:

Lobe-finned fish. Have muscular lobes associated with their fins and usually use lungs for gas exchange. Pneumatic sacs function as lungs or swim bladder. Paired fins with muscular lobes. Some bony fishes are hermaphrodites and a number of species exhibit parthenogenesis. Fertilization is usually external. Development is usually oviparous (egg laying) but can be ovoviviparous or viviparous.

Examples: Lungfishes, coelacanths

29. What is the scientific name of the shovel-nose sturgeon?

Scientific name of Shovelnose sturgeons is ***Scaphirhynchus platorynchus.***

30. Explain marine fisheries with examples.

Marine Fisheries:

Marine fisheries are conducted in all the oceans and seas of the world, including bays and estuaries. The capture and culture of aquatic organisms in salt water accounts for the bulk of the fishery products that reach world markets. Marine fishery deals with the fishery aspects of the sea water or ocean.

Example: Hilsa migrates from the sea to the river for breeding. Dr. Hora studied the migration of Hilsa.

31. What is Romer's classification?

Romer’s classification (1959):

Romer’s classified fishes into three classes.

- **Osteichthyes**: includes all bony fishes: **Teleostomi and Dipnoi**
- **Chondrichthyes**: includes all cartilaginous fishes
- **Placodermi**: including extinct fishes

32. Differentiate between a lagoon and a lake.

Lagoon:

- A lagoon is a shallow water body separated from the ocean by a barrier, such as a sandbar, reef, or barrier island.
- It typically contains saline or brackish water due to its connection with the ocean through narrow channels or inlets.
- Lagoons are dynamic ecosystems influenced by tides, waves, and coastal processes.
- Common examples include the Chilika Lake Lagoon in India and the Venice Lagoon in Italy.

Lake:

- A lake is an inland water body surrounded entirely by land and generally filled with freshwater, though some lakes may be saline (e.g., the Dead Sea).
- It has no direct connection to the ocean and is primarily fed by rivers, streams, or precipitation.
- Lakes are often stable ecosystems and can vary greatly in size and depth, from small ponds to large lakes like Lake Superior.

33. Name the pathogens and parasites that cause diseases in fish.

Fish Diseases Caused by Parasites and Pathogens:

1. **Bacterial Diseases:** Two bacterial diseases are very important.
   (i) Abdominal dropsy of Carps is caused by Aeromonas punctuata.
   (ii) Furunculoris of Salmons and trout's is caused by Aeromonas salminicida.

2. **Viral Diseases:** Economically most important is the viral haemorrhage septicaemia (VHS) of rainbow trouts.
3. **Protozoan Diseases:** Main protozoan diseases are caused by Costia, Myxobolus and Trypanosoma.
4. **Fungal Diseases**: The gill rot (branchyomyces) of carps involves the attack of Saprolegnia on the gills of carps.
5. **Worm Diseases**: Worms of four groups are parasites on fish. The flatworms (trematodes), tapeworms (cestodes), round worms (nematodes) and thorny-headed worms (acanthocephalans).
6. **Common Ectoparasites**: Two ectoparasites of fish are most important, the fish lice (Argulus, Lernaea and Ergasilus) and the fish leech (Piscicold). Both parasites weaken fish by feeding on its blood.

34. What is ciguatera poisoning?

Ciguatera poisoning:

Ciguatera poisoning is due to the consumption of fishes contaminated with ciguatoxin, a toxin which acts as a sodium channel agonist. This toxin is produced by a microscopic marine dinoflagellate Gambierdiscus toxicus and through the food chain it reaches the fish, mainly the reef fishes. The toxin accumulates mainly in the liver and testis of the fish as well as other viscera, while the flesh (muscle) of the fish is much less affected. Cooking or storage of the fish has no effect on the toxicity. Some of the fishes which have been implicated with ciguatera poisoning are groupers, barracuda, amberjack, sturgeon fish.

35. What is limnology? Write its scope.

Limnology is the study of the structural and functional interrelationships of organisms of inland waters as they are affected by their dynamic physical, chemical, and biotic environments.

Scope of Limnology:

1. **Physical Properties**: Studies temperature, light, currents, and water stratification, influencing aquatic organisms and ecosystem dynamics.
2. **Chemical Composition**: Examines water chemistry, nutrient cycles, and pollution, affecting water quality and ecosystem health.
3. **Biological Interactions**: Focuses on the relationships between aquatic organisms and their impact on the ecosystem.
4. **Water Quality and Pollution**: Assesses and manages the effects of pollutants on freshwater systems, aiming for water quality improvement.
5. **Conservation and Management**: Works on the sustainable management, restoration, and conservation of freshwater ecosystems.

36. How freshwater fishes regulate their internal salt concentration?

Freshwater fishes live in water with a much lower concentration of salts than they require inside their bodies. As a result, osmosis tends to promote the absorption of water into their bodies. To counteract this, their kidneys efficiently excrete water, and their gills reabsorb salt from the water. This helps maintain a stable internal environment and prevents excessive water uptake.

37. Why Actinopterygians are known as ray-finned fishes?

Their fins lack muscular lobes that is why called as ray-finned fishes. They usually possess swim bladders, gas filled sacs along dorsal wall of the cavity that regulate buoyancy. Paired fins supported by dermal rays. Base of fins is not muscular. Tail fins almost equal to upper and lower lobes. Blind olfactory.

Ray-finned fishes include herbivores, carnivores, omnivores, zooplankton (for example, small crustaceans such as copepods and fish larvae) consumers, and detritus feeders, or those that feed on dead organic matter.

38. Fish mouth morphology.

Mouth Type of fish:

- **Long and skinny bill:** Tweezer like, Poking into crevices.
- **Downward orientation:** Oriented mouth, Useful to suck food from bottom, E.g., Janitor fish
- **Large mouth:** Tearing large pr
- **Beaklike mouth:** used to graze on small algae growing on hard surfaces. E.g. Parrot fish
- **Upward orientation:** Used to capture prey on water surface.

Mouth Position of fish:

Most fish mouths fall into one of three general types:

1. **Superior**, or sometimes called **supra-terminal, mouths** are upturned. • The superior mouth is oriented upwards, and the lower jaw is longer than the upper jaw. • Archers, half-beaks, and hatchet fish are all examples of species of aquarium fish that have a superior mouth.
2. **Terminal mouths** point straight forward and are the most common mouth type. Terminal mouths are located in the middle of the head and point forward. Both jaws are the same length. More fish have this mouth type than any other.• Most barbs, cichlids, gouramis, and tetras have terminal mouths.
3. **Inferior**, **or sub-terminal, mouths** are turned downward. • Also called a sub-terminal or ventral mouth, the inferior mouth is turned downward. The lower jaw is shorter than the upper jaw, and the jaw will often be protrusible. • Most members of the catfish family have inferior jaws, and many of them also have a sucker mouth as well.
4. **Protrusible mouth** • A protrusible mouth allows a fish to extend its reach when attempting to snatch prey or food particles. • For example, kissing gourami uses its protrusible mouth to defend territory against others of the same specie.

39. Feeding behavior of fish. OR What prey do fish consume as food? OR Write the cassification of fish based on feeding habitat.

Fish consume a variety of prey depending on their species, habitat, and dietary preferences. Their diet can be broadly categorized as carnivorous, herbivorous, or omnivorous. Here are examples of prey fish commonly consume:

1. **Carnivores:** Fish exclusively consume animal matters as food.
2. **Herbivores:** Fish exclusively consume plant matters as food.
3. **Omnivores:** Fish consume both plant and animal matters as food.
4. **Larvivores:** Fish exclusively consume insect larvae as food.
5. **Planktivorous:** Fish exclusively consume plankton as food.
6. V**oracious**: Fish consuming or craving large quantities of food.
7. **Detritovores:** Fish consume decaying plants and animals.
8. **Cannibalistic:** Cannibalism is the act of consuming another individual of the same Species as food.
9. **Piscivores:** A piscivore is a carnivorous animal that eats primarily fish.

40. What is a food web?

A **food web** is a complex network of interconnected food chains within an ecosystem. It illustrates how different organisms are related to each other based on their feeding

relationships. At the base of the food web are **producers** like plants and algae, which convert solar energy into food through photosynthesis. These producers are consumed by **herbivores** (primary consumers), which are, in turn, eaten by **carnivores** (secondary and tertiary consumers), creating a flow of energy through various levels.

In addition to consumers, **decomposers** such as bacteria and fungi play a vital role in the food web by breaking down dead organic matter and recycling nutrients back into the environment. Unlike a simple food chain, a food web accounts for the complexity of real ecosystems, where organisms often occupy multiple feeding roles and interact with various species at different trophic levels. This interconnectedness makes food webs more resilient to changes in the ecosystem, as the loss of one species doesn't completely disrupt the flow of energy.

41. Write the importance of taxonomy and systematics.

Importance of Taxonomy and systematics:

Sound systematic data are important for conservation. Without accurate taxonomy,

- It is impossible to identify the species and evaluate their conservation status,
- It is impossible to properly manage their fisheries,
- It is impossible to evaluate the conservation value of habitats or areas,
- It is impossible to establish strategies
- It is impossible to set priorities.

42. What is the dorsal fin? Write its role in locomotion.

Dorsal Fins:

Dorsal fins are located on the back or on the top of the fish, and aid the fish in sharp turns or stops, and assist the fish in rolling. Fish may have up to three distinct dorsal fins, known as proximal, middle, and distal dorsal fins, however several fish have just two dorsal fins with the middle and distal fins fused together.

Fin Types:

The type of Doral fins are:

- Spiny and soft-rayed dorsal fin.
- Very long dorsal fin
- Tucked dorsal fin
- Locking spiny dorsal fin
- Three dorsal fins
- No dorsal fin

43. What are herring-like fishes?

Herring-like fish refer to species within the order **Clupeiformes**, which are small to medium-sized, schooling fish commonly found in marine and freshwater environments.

These fish are vital in marine food chains, serving as prey for larger fish, marine mammals, and birds.

Characteristics of Herring-like Fish:

1. **Examples**: Herring, sardines, shad, anchovies, and menhaden.
2. **Physical Traits**: Slender, streamlined bodies with silvery scales, adapted for fast swimming in schools.
3. **Habitat**: Found in oceans and coastal areas worldwide, with some species inhabiting freshwater or brackish waters.
4. **Diet**: Primarily plankton feeders, consuming small crustaceans, algae, and larvae.
5. **Ecological Importance**: Serve as a key food source for predators and contribute to nutrient cycling in aquatic ecosystems.
6. **Commercial Value**: Widely harvested for human consumption, fish oil, and fishmeal.

Herring-like fish play a crucial role in sustaining aquatic biodiversity and supporting fisheries globally.

44. How to reduce turbidity of pathogens and small particles?

To reduce turbidity caused by pathogens and small particles in water, a combination of physical, chemical, and biological treatment methods is used. Here are effective strategies:

1. **Coagulation and Flocculation**: Add chemicals like alum to bind small particles into larger flocs.
2. **Sedimentation**: Allow flocs to settle at the bottom of a tank for removal.
3. **Filtration**: Use sand, gravel, or membrane filters to remove small particles and pathogens.
4. **Disinfection**: Apply chlorine, ozone, or UV light to kill or inactivate remaining pathogens.
5. **Activated Carbon Filtration**: Absorb fine particles and some pathogens.
6. **Electrocoagulation**: Use electric current to aggregate and remove particles.
7. **Natural Methods**: Utilize wetlands or plant-based filters to reduce turbidity naturally.

45. What is the ecological role of cartilaginous fishes?

Cartilaginous fish, like sharks, rays, and chimaeras, play a vital role in the ecosystem. They live in various habitats and have diverse feeding habits. Some sharks are efficient predators, while others, like the whale shark, filter plankton. Rays and skates feed on crustaceans and mollusks using special teeth that crush tough shells.

These creatures inhabit both deep ocean and open water areas. Cartilaginous fish help maintain balance by controlling prey populations and supporting the health of marine ecosystems.Additionally, their diverse hunting and feeding strategies, such as fast-start predation, ram feeding, and sit-and-wait tactics, contribute to the intricate web of interactions in aquatic environments. These behaviors help regulate the populations of other marine species, ensuring the overall stability and health of the underwater ecosystems.

46. Fish challenges job today.

Challenges in the Fishing Industry:

1. **Impact of Overfishing:**
    - Reduced fish catch leads to job losses in fishing communities and decreases the food supply.
    - Progressive reduction in fish size increases the demand for food to meet nutritional needs.
    - Increased fishing efforts strain resources and equipment.
    - Bycatch (untargeted, protected, or endangered species) is harvested and often discarded, leading to waste and environmental damage.
    - Loss of marine biodiversity disrupts ecosystems and food chains.
2. **Farming Marine Species:**
    - The rising human population has created a demand for fish that exceeds the natural production capacity of oceans and seas.
    - Overexploitation of fisheries has caused a decline in wild fish populations.
    - Fish farming in controlled environments has become a solution to meet the growing demand while reducing pressure on wild fish stocks.

47. What is turbidity?

Turbidity is the measure of the cloudiness or haziness of a liquid, typically water, caused by the presence of suspended particles. These particles can include sediment, silt, algae, plankton, or organic and inorganic matter that scatter and absorb light, reducing water clarity.

1. **Causes**:
    - Soil erosion.
    - Algal blooms.
    - Waste discharge.
    - Urban runoff.
    - Natural disturbances in water bodies (e.g., storms, floods).
2. **Measurement**:
    - Measured in **NTU (Nephelometric Turbidity Units)** or **FTU (Formazin Turbidity Units)** using a turbidity meter.
3. **Impacts**:
    - Reduces light penetration, affecting aquatic plants' photosynthesis.
    - Can harm aquatic organisms by clogging gills or reducing oxygen levels.
    - Indicates possible pollution or contamination in water.
4. **Significance**:
    - Turbidity is an important parameter in water quality assessments for environmental, industrial, and drinking water standards.

48. What is a middle or marginal feeder?

Middle feeders (middle-water fish): Middle feeders (middle-water fish) often live in the middle layer of the aquarium and have small, forward-facing mouths. Because they eat feed that is placed directly in front of them. **Example:** Osteochilus thomassi

Marginal feeders: Marginal feeders prefers algae and other submerged vegetation, **a**nd is primarily a herbivorous column feeder**. Example:** Rohu

49. What is a voracious or cannibalistic feeder?

Cannibalistic:

Cannibalism is the act or behavior in which an individual consumes another individual of the same species as food. This behavior can occur for various reasons, including a lack of other food sources, dominance, territorial disputes, or as a survival strategy. In some species, cannibalism is a common practice, especially when food is scarce or in environments where individuals are isolated. It can also be seen in certain predator-prey dynamics, where stronger individuals prey on weaker or younger members of the same species. This behavior may also serve to reduce competition or eliminate potential rivals. In some cases, cannibalism can be a natural part of the life cycle, such as in some amphibians, fish, or insects.

Voracious:

Fish consuming or craving large quantities of food. Having or marked by an insatiable appetite for an activity or pursuit; greedy. Voracious predators sucking body fluids of many insects or blood of fish, tadpoles and frogs. Recommended food for the voracious would be:

• Shark is an active, voracious predator of fish and invertebrates.

• Eels are various carnivores and consume a variety of fishes.

• and invertebrates such as worms, crayfish, snails and insects.

Examples: Piranhas, Barracuda, Pike.

50. Write the function of the pyloric ceca and stomach in the digestion of fish.

Function of the Pyloric Ceca and Stomach in the Digestion of Fish:

1. **Stomach**
   - The **stomach** structure varies greatly among fish, depending on their diet.
   - In most **predatory fish**, it is a simple straight or curved tube or pouch with a muscular wall and glandular lining.
   - Food is primarily digested here, leaving the stomach in a **liquid form**.
   - The junction between the stomach and the intestine is marked by a **muscular valve**.
2. **Pyloric Ceca**
   - The **pyloric ceca** are blind sacs located at the junction between the stomach and intestine in some fish.
   - These structures perform **digestive, absorptive**, or both functions, aiding in nutrient breakdown and assimilation.

51. What is composite fish farming?

Composite Fish Farming:

It is found that if few selected species of fish are stocked together in proper proportion in a pond, total production of fish is increased many times. This mixed farming is called composite farming. It has some advantage-compatible species do not harm each other, all available areas are fully utilised, no competition among different species is found and fish may have beneficial effect on each other. Catla catla, Labeo-rohita and Cirrhina mrigala are surface feeder, column feeder and bottom feeder respectively and are used for composite farming.

52. What is the impact of the fishing industry on farmers and the country?

Impact of the Fishing Industry on Farmers and the Country

1. **Employment Generation**
    - The development of the **fishing industry** creates numerous employment opportunities, providing livelihoods to millions, particularly in coastal and rural areas.
2. **Source of Income**
    - The fishing industry serves as a significant **source of income** for farmers and contributes substantially to the national economy.
    - It enhances the financial stability of farmers while boosting export revenues for the country.
3. **Blue Revolution**
    - The rise in **fish production** has led to the concept of the "Blue Revolution," on the same line as "Green Revolution" in agriculture, signifying the industry's role in ensuring food security and economic growth.

53. Write the names of important edible fishes.

Some important edible fish are:

- Rohu (Labeo rohita)
- Calbasu (L. calbasu)
- Catla (Catla catla)
- Singhara (Mystus seenghala)
- Mangur (Clarias batrachus)
- Singhi (Heteropneustes fossilis)
- Malli (Wallago attu)
- Mirgala (Cirrhinus mrigala)

54. What are detritivores?

Detritivores are organisms that feed on decaying organic matter, such as dead plants and animals, as their primary source of nutrition. Around **10 percent of all fish species** are

detritivores, consuming decomposing matter to obtain energy. In addition to decaying matter, some detritivores may practice **coprophagy**, which involves eating feces.

Examples of detritivores include:

- **Lobsters**
- **Fiddler crabs**

In aquatic ecosystems, detritivorous fish play a significant role in nutrient cycling. For instance, in the **Amazon basin**, detritivorous fish, primarily from the orders **Characiformes** and **Siluriformes** (catfishes), form an important part of the ichthyomass. Characiformes contribute more than **30%** of the total fish yield, with Siluriformes being of minor importance. These fish obtain their carbon through food chains originating from **phytoplankton** or other plant sources.

Marine detritivores, like **crabs**, **lobsters**, and **sea cucumbers**, are often found on or within the seabed, known as the **benthos**. These organisms are commonly referred to as **"bottom-feeders"**. Some aquatic organisms, such as **polychaete worms**, **barnacles**, and **corals**, use **filter feeding** to obtain energy from floating organic detritus known as "marine snow."

55. What type of food is given to carnivorous fish?

Carnivorous:

Carnivorous fish that eat things made from animals such as aquatic invertebrates and vertebratebts (fish larvae, fry and small fish mollusks, crustaceans) are called carnivorous fish. Such fish are usually predatory.

Recommended food for the carnivores would be:

- Carnivores are happiest when they are fed live food like worms.
- Earthworms
- Red worms
- Daphnia
- Tubifex worms
- Shrimps
- Oysters
- Clams and other fish.
- Lean chicken and salmon. These should be cooked, but never fried.

56. What is PUFA?

PUFA stands for **Polyunsaturated Fatty Acids**, which are a type of healthy dietary fat. These fats have more than one double bond in their chemical structure, making them "polyunsaturated." PUFAs are essential for the body because they cannot be synthesized internally and must be obtained through the diet.

Key Types of PUFA:

1. **Omega-3 Fatty Acids** (e.g., EPA, DHA, ALA): Found in fish (like salmon and mackerel), flaxseeds, chia seeds, and walnuts.
2. **Omega-6 Fatty Acids** (e.g., linoleic acid): Found in vegetable oils (like sunflower, soybean, and corn oils) and nuts.

Functions of PUFA:

- Support brain function and development.
- Reduce inflammation.
- Lower bad cholesterol levels and improve heart health.
- Maintain healthy skin and cell membranes.

A balanced intake of omega-3 and omega-6 PUFAs is crucial for optimal health.

57. What is an omnivorous fish?

Omnivores:

Fish consume both plant and animal matters as food. Omnivore fish are destructive to plants and other small animals because they will feed just about anything. Additionally, they are voracious eaters. These animals frequently exhibit the urge to overfeed, and when they do, the fat tends to accumulate quickly. The mouth is smaller and is usually devoid of teeth except for pharyngeal teeth that may be blunt and flat (molariform) for grinding or sharp and long for shredding.

This group of fish ingests both higher proteins and vegetable matter. They also comprise the easiest fish to feed as they like almost any type of food; flake foods, live foods and pretty much anything else. With the most versatile stomachs, omnivores' digestive tracts can derive nutrients from a variety of dietary options.

58. Differentiate between marine and freshwater ecosystems.

Freshwater Ecosystem:

The freshwater ecosystem is an aquatic ecosystem that includes lakes, ponds, rivers, streams and wetlands. These have no salt content in contrast with the marine ecosystem. The freshwater ecosystem is mainly divided into three types based on its region: Lotic, lentic, and wetland freshwater ecosystem.

Marine ecosystem:

The marine ecosystem includes seas and oceans. These have a more substantial salt content and greater biodiversity in comparison to the freshwater ecosystem. A marine ecosystem is a dynamic and interconnected community of living organisms, including tiny plankton, massive blue whales, and everything in between. These ecosystems cover over 70% of the Earth's surface and encompass a vast range of habitats, from shallow tide pools to the deepest ocean trenches. Marine ecosystems provide essential services, including primary production, food webs, nutrient cycling, coastal protection, and climate regulation, supporting an incredible array of biodiversity. They are crucial for human well-being, providing half of the oxygen we breathe, a significant source of food, and livelihoods for millions of people.

59. What is the telencephalon and diencephalon?

Telencephalon and Diencephalon:

In fish brains, the brain is generally divided into three regions: **Prosencephalon**, **Mesencephalon**, and **Rhombencephalon**. The **prosencephalon** gives rise to two important parts:

1. **Telencephalon**
    - The **telencephalon** is the **anterior part** of the brain, primarily responsible for the reception and conduction of **smell**.
    - It is divided into two regions: **Area ventralis** and **Area dorsalis**.
    - The **ventral area** receives secondary olfactory fibers and is a constant structure in all **bony fishes**.
    - In adult fishes, the telencephalon may form through **invagination and inversion**, and plays a crucial role in **reproductive behavior**.
2. **Diencephalon**
    - The **diencephalon** is primarily concerned with **vision** and is divided into three parts:
      (i) **Epithalamus**
      (ii) **Hypothalamus**
      (iii) **Thalamus**

60. Classification of Carp.

Classification of Carp:

- Domain: Eukarya
- **Kingdom**: Animalia
- **Phylum**: Chordata
- **Class**: Actinopterygii
- **Order**: Cypriniformes
- **Family**: Cyprinidae
- **Genus**: Cyprinus
- **Species**: Cyprinus carpio (Common Carp)

61. What is Caudal Peduncle?

The **caudal peduncle** is the base of the caudal fin.

Peduncle means stem, and the caudal peduncle is where the strong swimming muscles of the tail are found. Together, the caudal fin acts like a “propeller” for the fish, and the caudal peduncle acts like a motor.

62. What is a limiting factor?

A **limiting factor** is the resource or condition that is in the shortest supply relative to the needs of an organism or process, thereby restricting the growth, development, or performance

of that organism or process. When this factor is scarce, it becomes the bottleneck, limiting the overall rate or extent of the process.

For fish, common limiting factors include oxygen levels in water, temperature, availability of food, and suitable habitat for spawning and shelter. Changes in water quality, pollution, or overfishing can exacerbate these limitations, significantly impacting fish populations. Understanding and managing these factors are essential for maintaining aquatic ecosystems and ensuring the sustainability of fish species.

63. What is gill protection adaptation?

Gill protection adaptation refers to the structural features and mechanisms that protect the delicate gill filaments in fish while allowing efficient respiration.

1. **In Teleosts and Many Other Fishes**: Gills are protected by a **gill cover (operculum)** on both sides of the pharynx. This bony structure shields the gills from physical damage and regulates water flow for respiration.
2. **In Sharks, Rays, and Some Fossil Fish Groups**: Gills are protected by **flaps of skin** rather than a bony cover. These skin flaps guard the gill slits and ensure protection while allowing water to pass through for breathing.

This adaptation varies across fish groups to suit their ecological and evolutionary needs.

64. What is Holocephaly?

Holocephaly refers to a subclass of **cartilaginous fish** within the class **Chondrichthyes**. These fishes are also known as **"complete heads"** (Holocephali) or sometimes referred to as **Euchondrocephali**. They are primarily characterized by the following features:

- **Operculum** covers the gill slits.
- Possess **crushing teeth**.
- **No scales** on their bodies.
- They have a **lateral line system**.
- Holocephali are known for their **large pectoral fins** which they use for sweeping movements to navigate.
- They are typically **deep-sea fish** with **slender tails**, living close to the seabed and feeding on **benthic invertebrates**.
- These fish lack a **stomach**, and their food moves directly into the intestine.

The earliest fossils of Holocephali, primarily teeth, date back to the **Devonian period**, but little is known about these primitive forms.

Example: **Ratfish**.

65. What are the challenges in enforcing fish regulations internationally?

Challenges in Enforcing Fish Regulations Internationally:

1. **Lack of Global Coordination**
    - Different countries have varied laws, policies, and enforcement mechanisms, creating difficulties in ensuring a unified approach to sustainable fishing practices.
2. **Illegal, Unreported, and Unregulated (IUU) Fishing**
    - IUU fishing undermines international regulations by allowing unregulated or illegal fishing activities that go unchecked, leading to overfishing and ecosystem damage.
3. **Overfishing and Resource Depletion**
    - In some regions, nations prioritize immediate economic gains over long-term sustainability, leading to overfishing and the depletion of fish stocks, even when international regulations are in place.
4. **Weak Enforcement and Monitoring**
    - Many countries lack the resources or infrastructure needed to effectively monitor fishing activities and enforce regulations, especially in remote or international waters.
5. **Economic and Political Pressures**
    - Fishing industries and political considerations can hinder stricter enforcement of regulations, as economic interests often conflict with the need for sustainable fishing practices.

Regards:

★ AlifExpertZoologist ★